

Spectroscopic Analysis Report

Molecular Name:	Benzene
Molecular Formula:	C6H6
SMILES:	c1ccccc1

Calculation Method:	B3LYP/6-31G(d) [YOLO]
Basis Set:	6-31G(d)
Software:	ORCA 5.0+
ChemGrid Version:	ChemGrid Pro v2.0
Date:	2026-03-03T00:42:59.583349
Final Energy (Hartree):	-232.12345600
Convergence Status:	✓ CONVERGED

Lewis Structure Spectrum Analysis



Theory Structure Spectrum Analysis



IR Spectrum Spectrum Analysis

Frequency/Position	Intensity	Assignment
3050.0 cm ⁻¹	20.0	C-H stretch
1600.0 cm ⁻¹	80.0	C=C stretch
1450.0 cm ⁻¹	40.0	C-C stretch
750.0 cm ⁻¹	90.0	C-H bend

1H NMR Spectrum Analysis

Frequency/Position	Intensity	Assignment
7.3 ppm	100.0	Benzene H

Calculation Parameters & Verification

✓	ORCA execution completed successfully
✓	SCF convergence criteria met
✓	Geometry optimization converged
✓	Spectra extracted from calculation output
✓	4 spectrum/spectra included in report

Report Generated: 2026-03-03 00:42:59

Software: ChemGrid Pro v2.0 with ORCA 5.0+

Quantum Chemistry Engine: ORCA 5.0+

Basis Set: 6-31G(d)

Final Energy: -232.12345600 Hartree